

Day 2: Bren Math Workshop

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The exponential function

The number e

The **number e is a constant** in mathematics, it is approximately 2.72, but really its decimals continue infinitely without any pattern:

$$e = 2.71828182845904523536...$$

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It's exact value can be described as

$$e = 1 + 1 + \frac{1}{1 * 2} + \frac{1}{1 * 2 * 3} + \frac{1}{1 * 2 * 3 * 4} + \frac{1}{1 * 2 * 3 * 4 * 5} + \dots$$

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Like any other number, e follows normal power properties:

$$\begin{array}{ll} e^x \cdot e^y = & ? \quad e^{x+y} \\ \frac{1}{e^x} = & ? \quad e^{-x} \\ \frac{e^x}{e^y} = & ? \quad e^{x-y} \\ (e^x)^r = & ? \quad e^{xr} \end{array}$$

 Take a moment to write these with a single exponent.

The number e

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Like any other number, e follows normal power properties:

$$e^{x+y} = e^x \cdot e^y$$

$$e^{-x} = \frac{1}{e^x}$$

$$e^{x-y} = \frac{e^x}{e^y}$$

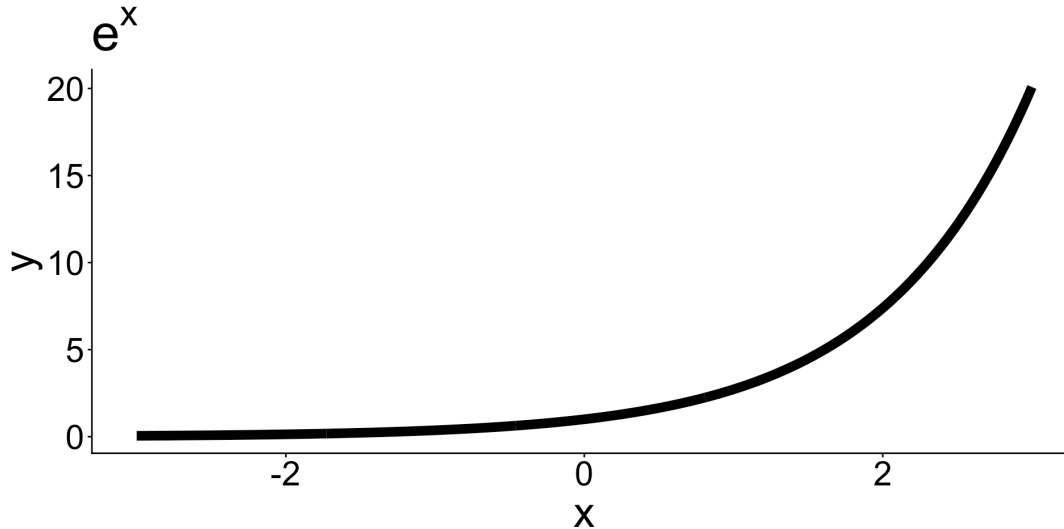
$$e^{rx} = (e^x)^r$$

The exponential function

The **exponential function** e^x is the number e raised to the x power.

In function notation you may sometimes see it as

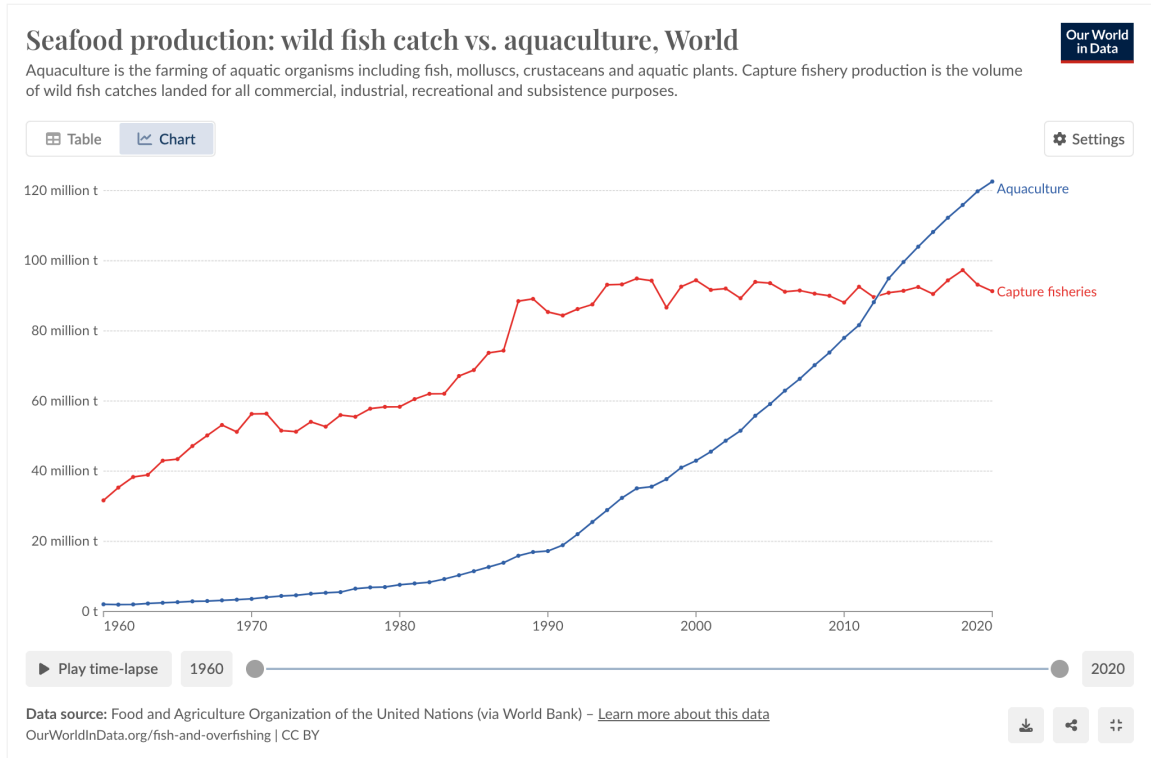
$$f(x) = e^x \text{ or } \exp(x).$$



Why is the exponential function so common?

- **One reason:** Exponential trends show up a LOT in environmental science (the proportional change is the same over each time span)
- **Math reason:** Turns out it's a very useful value for calculus

A real-world example:



Logarithms

Logarithms

$$\log_a(b)$$

$\log_a(b)$ is the exponent you need to raise a to get a value of b

For example:

- $\log_2(8) = x$ asks “to what power do I have to raise 2, to get a value of 8?”
- $\log_{105}(1) = x$ asks “to what power do I have to raise 105, to get a value of 1?”

$\log_2(8) = 3$ because $2^3 = 8$, so 3 is the exponent I need to raise 2 to to get 8.

$\log_{105}(1) = 0$ because $105^0 = 1$.

Natural logarithms



Natural logarithms

If $y > 0$, then $\ln(x)$ is the exponent you need to raise e to in order to get y .

This means:

$$\ln(e^x) = x$$

$$e^{\ln(y)} = y$$

$$\ln(e^x) = x$$

$$e^{\ln(y)} = y.$$

In other words, $\ln(x)$ is the inverse of the exponential function e^x .

Examples

- $\ln(e^3) = 3$, bc 3 is the power of e needed to get e^3 .
- $\ln(\frac{1}{e}) = \ln(e^{-1}) = -1$
- $\ln(1) = 0$, because $e^0 = 1$.
- $\ln(0)$ means I can raise e to the something to get 0.
- $\ln(-7)$

Natural logarithms

If $y > 0$, then $\ln(x)$ **is the exponent you need to raise e to in order to get y .**

This means:

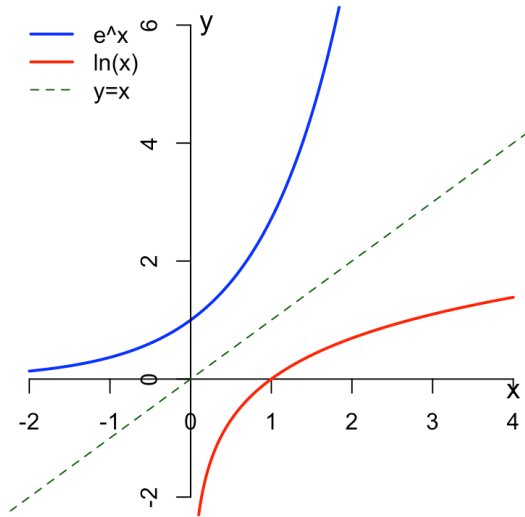
$$\begin{aligned}\ln(e^x) &= x \\ e^{\ln(y)} &= y\end{aligned}$$

In other words, $\ln(x)$ is the inverse of the exponential function e^x .

Examples

- $\ln(e^3) = 3$ because 3 is the power of e needed to get e^3 .
- $\ln(\frac{1}{e}) = \ln(e^{-1}) = -1$
- $\ln(1) = 0$ because 0 is the exponent we need to raise e to to get 1
- $\ln(0)$ does not exist, because there's no number we can raise e to to get 0
- $\ln(-7)$ does not exist, because e to any number always be positive

Natural log properties



$$\ln(e^x) = x$$

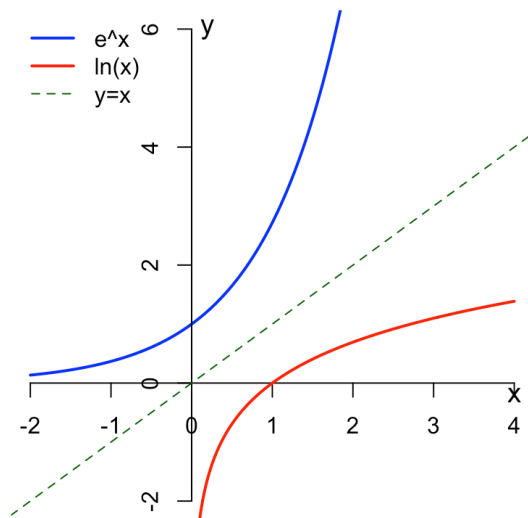
$$\ln(xy) = \ln(x) + \ln(y)$$

$$\ln\left(\frac{x}{y}\right) = \ln(x) - \ln(y)$$

$$\ln(x^y) = y \cdot \ln(x)$$

Natural log properties

Properties



$$\ln(e^x) = x$$

$$\ln(xy) = \ln x + \ln y$$

$$\ln\left(\frac{x}{y}\right) = \ln x - \ln y$$

$$\ln(x^y) = y \ln x$$

Using logarithms to solve equations

 Solve for t in the equation $Pe^{rt} = A$.

$$Pe^{rt} = A$$

$$e^{rt} = \frac{A}{P}$$

$$\ln(e^{rt}) = \ln\left(\frac{A}{P}\right)$$

$$rt = \ln\left(\frac{A}{P}\right)$$

$$t = \frac{1}{r} \ln\left(\frac{A}{P}\right) = \frac{\ln(A) - \ln(P)}{r}.$$

Properties

- $\ln(e^x) = x$
- $\ln(xy) = \ln x + \ln y$
- $\ln\left(\frac{x}{y}\right) = \ln x - \ln y$
- $\ln(x^y) = y \ln x$

Using logarithms to solve equations

 Solve for t in the equation $Pe^{rt} = A$.

$$Pe^{rt} = A$$

$$e^{rt} = \frac{A}{P}$$

$$rt = \ln\left(\frac{A}{P}\right)$$

$$t = \frac{\ln(A) - \ln(P)}{r}$$

Applications

Exponential growth

When a population grows at a rate that is directly proportional to its size, we can model the population size at a time t as

$$N(t) = N_0 e^{rt}$$

where

- r = per capita intrinsic growth rate
- N_0 = initial population size = population at $t = 0$



Take a minute to solve these individually.

1. What per capita intrinsic growth rate would make the population remain stable across time?

2. Are the following statements true or false? Explain your reasoning.

- If $r = \ln(\frac{1}{2})$, then the population will be half of its original size at $t = 2$.
- If $r = \ln(4)$, then the population will be four times its original size at $t = 1$.

3. Make a hypothesis about how r being positive or negative is related to growth or decay of $N(t)$.
4. Assuming $r > 0$, when will the population double?

$$N(t) = N_0 e^{rt}$$

Exponential growth practice

1. What per capita intrinsic growth rate would make the population remain stable across time?

2. Are the following statements true or false? Explain your reasoning.

• If $r = \ln(\frac{1}{2})$, then the population will be half of its original size at $t = 2$.

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I need to calculate $N(2)$ where $r = \ln(\frac{1}{2})$.

 Let's see a solution!

$$N(t) = e^{rt} \cdot N_0 \rightsquigarrow N(1) = e^{\ln(4) \cdot 1} \cdot N_0 = e^{\ln(4)} \cdot N_0 = 4N_0.$$

$$\Rightarrow N(1) = 4N_0.$$

$$N(t) = N_0 \cdot e^{rt} \rightsquigarrow N(2) = N_0 \cdot e^{\ln(\frac{1}{2}) \cdot 2} = N_0 (e^{\ln(\frac{1}{2})})^2$$

$$\Rightarrow N(2) = \frac{N_0}{4}.$$

$$e^{ab} = (e^a)^b$$

$$= N_0 \left(\frac{1}{2}\right)^2 = N_0 \cdot \frac{1}{4}.$$

Exponential growth practice

3. Make a hypothesis about how r being positive or negative is related to growth or decay of $N(t)$.
4. Assuming $r > 0$, when will the population double?

 Let's see a solution!

$$N(t) = N_0 \cdot e^{rt} \quad , \text{ pop doubling: } 2N_0$$

$$e^{rt} = (e^r)^t$$

$$\ln(e^{rt}) = \ln((e^r)^t)$$

$$= t \cdot \ln(e^r)$$

$$= t \cdot r$$

$$\text{Solve for } t: 2N_0 = N_0 \cdot e^{rt} \Rightarrow 2 = e^{rt}$$

$$\Rightarrow \ln(2) = \ln(e^{rt})$$

$$\Rightarrow \ln(2) = rt$$

$$\Rightarrow \frac{\ln(2)}{r} = t$$

$$r = -2$$

$$t = 2$$

$$N(2) = N_0 \cdot e^{(-2)(2)}$$

$$= N_0 \cdot e^{-4}$$

Solutions

$$x^{ab} = (x^a)^b \neq x^{(a^b)} \quad \text{incorrect.}$$
$$2^{(-1)(2)} = 2^{-2} = \frac{1}{4} \quad // \quad 2^{(-1)^2} = 2^1 = 2.$$

(1) A stable population means $N(t) = N_0$ for every t , so we need $e^{rt} = 1$ for all t , which requires $r = 0$.

(2)

- *False.* $N(2) = N_0 e^{2 \ln(1/2)} = N_0 \left(\frac{1}{2}\right)^2 = \frac{N_0}{4}$, a **quarter** of N_0 , not a half.
- *True.* $N(1) = N_0 e^{\ln(4)} = N_0 \cdot 4 = 4N_0$.

(3) The sign of r determines whether $N(t)$ **grows** ($r > 0$) or **decays** ($r < 0$). See the plot on the next slide!

(4) Solve $N_0 e^{rt} = 2N_0$ for t :

$$e^{rt} = 2 \Rightarrow rt = \ln(2) \Rightarrow t = \frac{\ln(2)}{r}$$

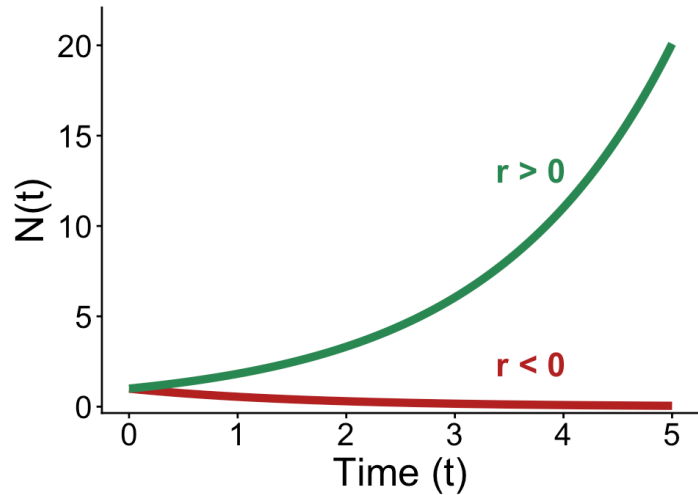
Exponential growth

When a population grows at a rate that is directly proportional to its size, we can model the population size at a time t as

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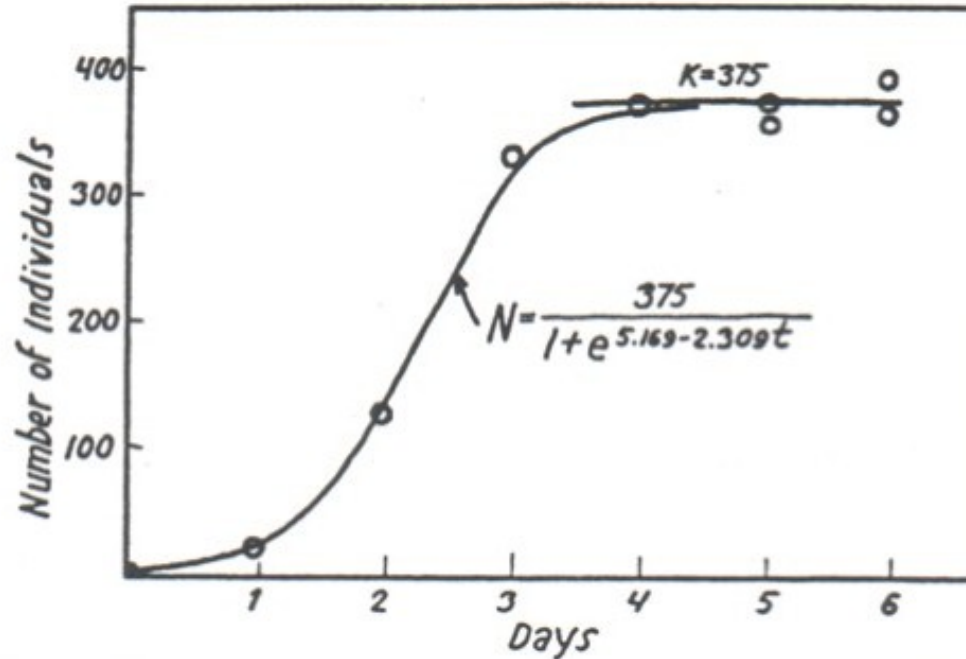
where

- r = per capita intrinsic growth rate
- N_0 = initial population size = population at $t = 0$



What environmental constraints could cause the population to not follow exponential growth?

But populations can't grow exponentially forever



The growth of population of *Paramecium caudatum*

Gause, G. F. 1934. The Struggle for Existence. Baltimore: Williams and Wilkins.

Let's take a 10 minute break



image: Flaticon.com

Linear functions

Linear functions: slope-intercept form

 Describes linear relationship between two variables

x and y .

$$\underbrace{y}_{\substack{\text{output} \\ \text{dependent variable}}} = \underbrace{m}_{\substack{\text{slope} \\ \text{input} \\ \text{independent variable}}} x + \underbrace{b}_{\substack{\text{y-intercept}}}$$

Remember! y -intercept is the y value we get when $x=0$
 $y = m \cdot 0 + b = \underline{\underline{b}}$

Linear functions: slope-intercept form

Easiest model to describe linear relationship between two variables!

$$\underbrace{y}_{\text{output}} = \underbrace{\widehat{m}}_{\text{Slope}} \underbrace{x}_{\text{input}} + \underbrace{\widehat{b}}_{y \text{ intercept}}$$

Finding the equation of a line

✎ Given two points (x_1, y_1) and (x_2, y_2) on the line:

1) Find m , the slope, by using the points:

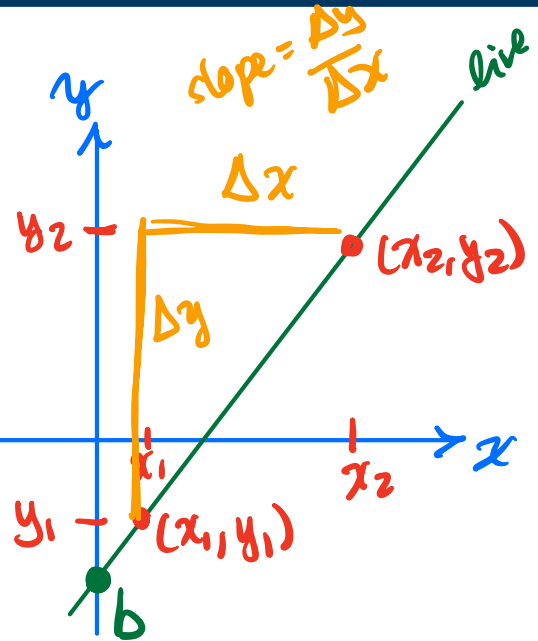
delta $\rightarrow m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{"rise"}}{\text{"run"}}$

2) Find b , the y -intercept.

Since (x_1, y_1) is on the line, we have

$$y_1 = mx_1 + b$$

Solve for b from here: $b = y_1 - mx_1$.



Finding the equation of a line

Given two points (x_1, y_1) and (x_2, y_2) on the line:

1. Find m , the **slope**, by using the two points:

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}.$$

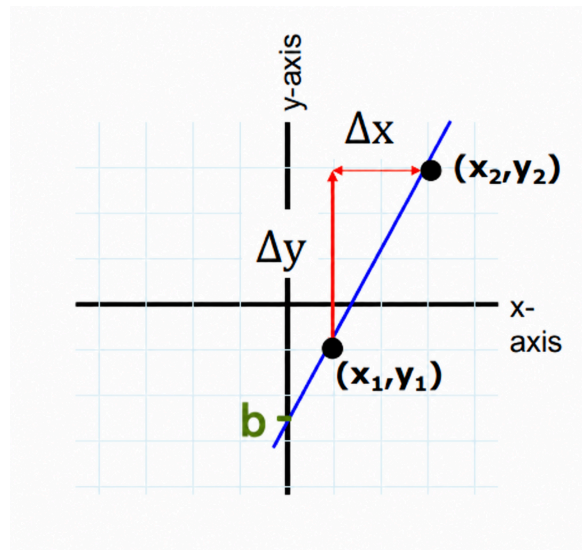
2. Find b , the **y-intercept**:

Since (x_1, y_1) is on the line, it satisfies:

$$y_1 = mx_1 + b.$$

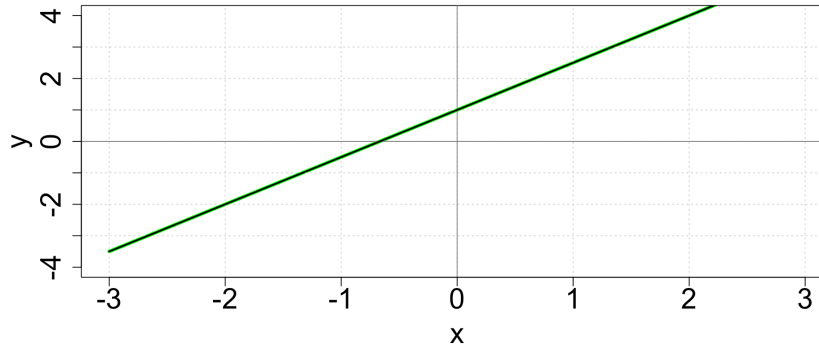
We can solve for b to obtain:

$$b = y_1 - mx_1.$$



Equation of a line: practice

- Find the equation of the line that:
 - a. passes through $(3, 5)$ and $(2, -1)$.
 - b. passes through $(2, -5)$ and has x -intercept equal to 4.
 - c. has the following graph:



- Make a graph showing the line that passes through $(3, -1)$ and has slope equal to 2.

Equation of a line: practice

A. Find the equation of the line that passes through $(3, 5)$ and $(2, -1)$.

 Let's see a solution.

x_1 y_1 x_2 y_2

Arrows pointing from x_1 to 3, y_1 to 5, x_2 to 2, and y_2 to -1.

$$\frac{5 - (-1)}{3 - 2} = \frac{6}{1} = 6 \Leftarrow \underline{\underline{m}} \text{ (slope)}$$

$$y = mx + b$$

$$5 = 6(3) + b$$

$$5 = 18 + b$$

$$\boxed{-13 = b}$$

$$\boxed{y = 6x - 13}$$

Equation of a line: practice

B. Find the equation of the line that passes through $(2, -5)$ and has x-intercept equal to 4.

 Let's see a solution.

$$\frac{0 - (-5)}{4 - 2} = \frac{5}{2} = 2.5 \rightarrow m \text{ (slope)}$$

$\rightarrow (4, 0)$

$$y = mx + b$$

$$0 = 2.5(4) + b$$

$$0 = 10 + b$$

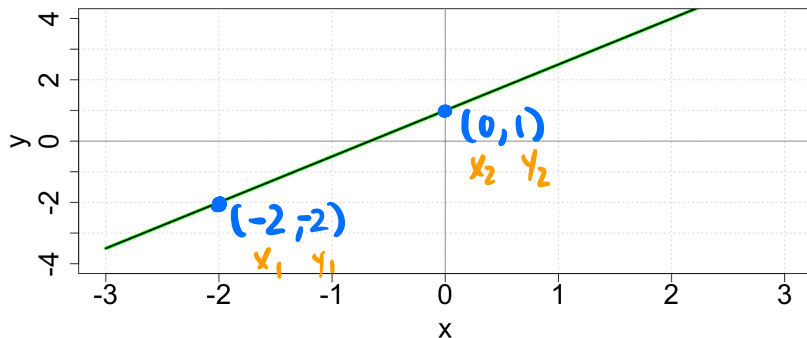
$$-10 = b \rightarrow y\text{-int}$$

$$\boxed{y = 2.5x - 10}$$

Equation of a line: practice

C. Find the equation of the line that has the following graph:

 Let's see a solution.



$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{1 + 2}{0 + 2} = \boxed{\frac{3}{2} = m}$$

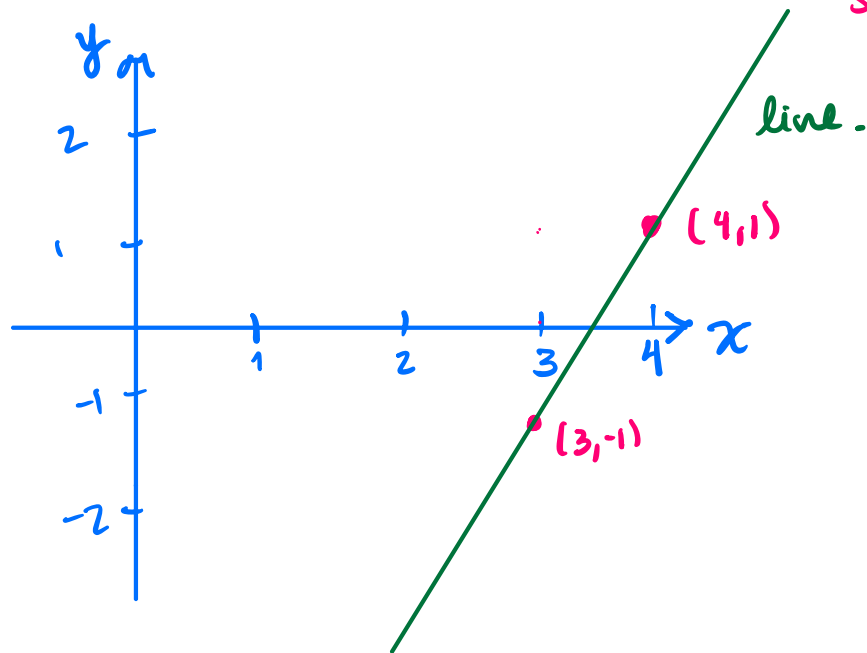
$$b = y_1 - m \cdot x_1$$

$$b = 1 - \left(\frac{3}{2}\right)0$$

$$\boxed{b = 1}$$

$$\boxed{y = \frac{3}{2}x + 1}$$

Make a graph for the line that passes through
 $(3, -1)$ and has slope $= 2$.



$$\text{slope} = 2 = \frac{2}{1} = \frac{\text{"rise"}}{\text{"run"}}$$

Solutions

a. Passes through (3,5) and (2,-1):

$$m = \frac{5 - (-1)}{3 - 2} = 6$$

Compute the slope

$$y - 5 = 6(x - 3)$$

Point-slope form using (3,5)

$$y = 6x - 13$$

Simplify

b. Passes through (2,-5) and has x-intercept equal to 4:

$$m = \frac{0 - (-5)}{4 - 2} = \frac{5}{2}$$

Compute the slope using the x-intercept (4,0)

$$y - 0 = \frac{5}{2}(x - 4)$$

Point-slope form

$$y = \frac{5}{2}x - 10$$

Simplify

c. Has the given graph: $y = \frac{3}{2}x + 1$

$$\text{slope} = \frac{\text{rise}}{\text{run}} = \frac{3}{2}$$

Read from the graph

$$\text{y-intercept} = 1$$

Read from the graph

$$y = \frac{3}{2}x + 1$$

Equation of the line

Solutions

Make a graph showing the line that passes through (3,-1) and has slope equal to 2.

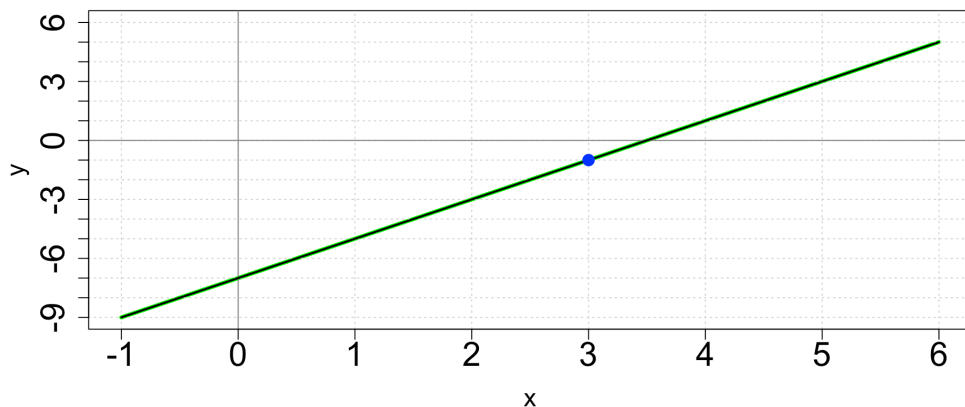
 Let's see a solution.

$$y - (-1) = 2(x - 3)$$

Point-slope form

$$y = 2x - 7$$

Simplify



Applications

Supply and demand

A regional market for albacore tuna is described below, where Q is quantity in thousands of pounds sold per week and P is price in \$ per pound.

 Take a minute to solve these individually.

1. A fishery's weekly supply curve for tuna passes through $(Q, P) = (0, 20)$ and $(100, 70)$. Find the supply curve $S(Q)$ in slope-intercept form.
2. At a price of \$70/lb, weekly quantity demanded is 40 (thousand lbs); at \$20/lb, quantity demanded is 140 (thousand lbs). Find the inverse demand equation $D(Q)$.
3. Plot $S(Q)$ and $D(Q)$ on the same axes, with quantity on the x -axis and price on the y -axis. Label both axes clearly, including units.
4. Find the price and quantity at which supply equals demand and interpret this point in one sentence.

Supply and demand

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Let's see a solution.

Supply and demand

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 Let's see a solution.

Supply and demand

4. Find the price and quantity at which supply equals demand and interpret this point in one sentence.

 Let's see a solution.

Solutions

1. Supply passes through $(0, 20)$ and $(100, 70)$:

$$m = \frac{70 - 20}{100 - 0} = 0.5 \quad \text{Compute the slope}$$
$$S(Q) = 20 + 0.5Q \quad \text{Use the y-intercept } (0, 20)$$

2. Demand passes through $(Q, P) = (40, 70)$ and $(140, 20)$:

$$m = \frac{20 - 70}{140 - 40} = -0.5 \quad \text{Compute the slope}$$
$$P = -0.5Q + c \Rightarrow 70 = -0.5(40) + c \Rightarrow c = 90 \quad \text{Using } (40, 70)$$
$$D(Q) = 90 - 0.5Q \quad \text{Simplify}$$

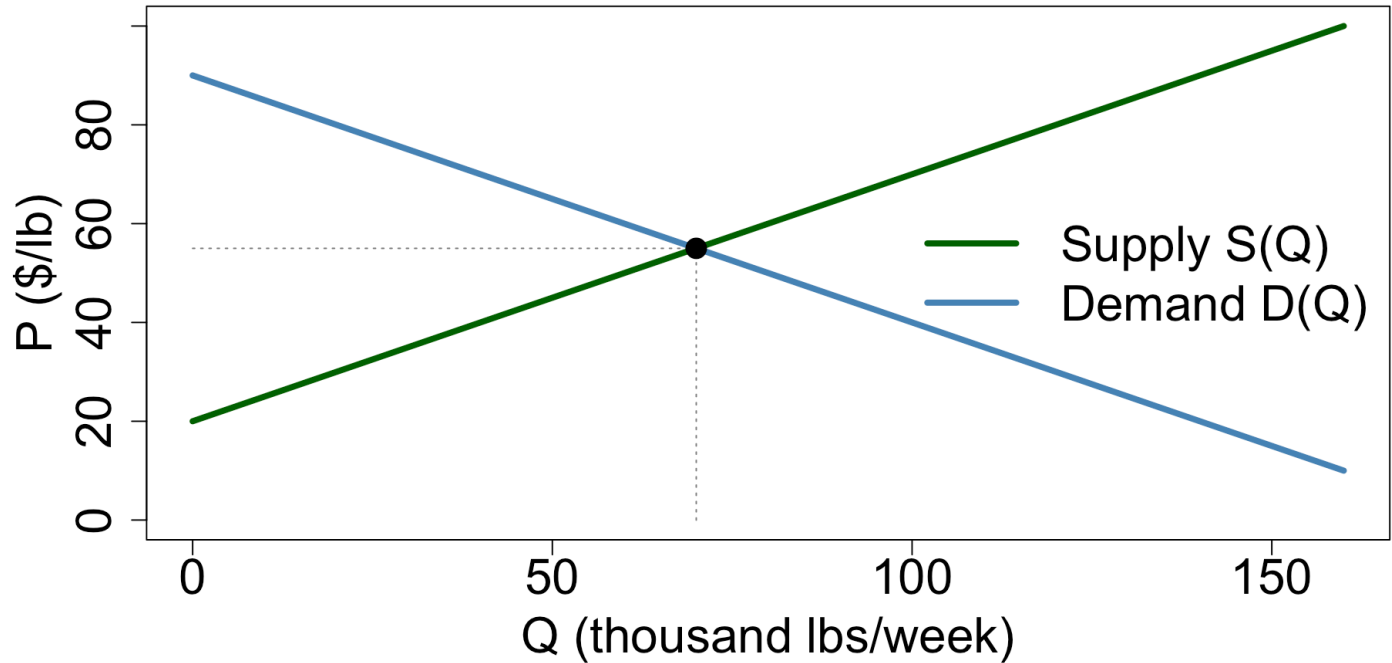
4. Set $S(Q) = D(Q)$:

$$20 + 0.5Q = 90 - 0.5Q \quad \text{Set supply equal to demand}$$
$$Q = 70 \quad \text{Solve for } Q$$
$$P = 20 + 0.5(70) = 55 \quad \text{Plug into either curve}$$

At a price of \$55/lb, fishers supply exactly 70 (thousand lbs/week) of tuna and buyers demand exactly that same amount.

Solutions

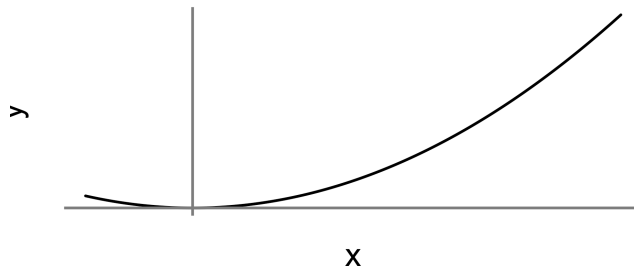
3. $S(Q)$ and $D(Q)$ plotted together:



Interpreting the slope

Slope (average)

Sometimes, it can be useful to find the **average rate of change** of a function.



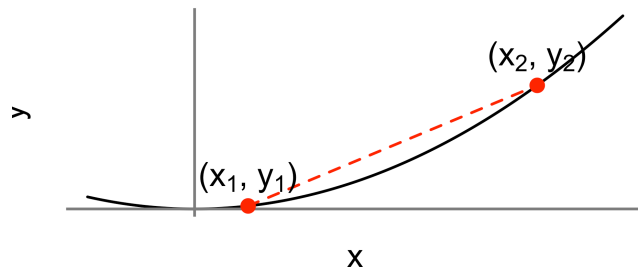
Slope (average)

Sometimes, it can be useful to find the **average rate of change** of a function.

Between any two points (x_1, y_1) and (x_2, y_2) on the graph of a function, the slope is found by:

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Same idea as in a line!



Slope represents rate of change

Ice-core data before 1958. Mauna Loa data after 1958.

Source: [The Keeling Curve](#)

Get into the practice of saying the meaning out loud

- As if you're explaining it to someone unfamiliar with the data
- Including units
- Without overstating certainty

For example:

“Between 1972 and 2020 the price of hobbit homes increased by an average of \$2,450 per year”

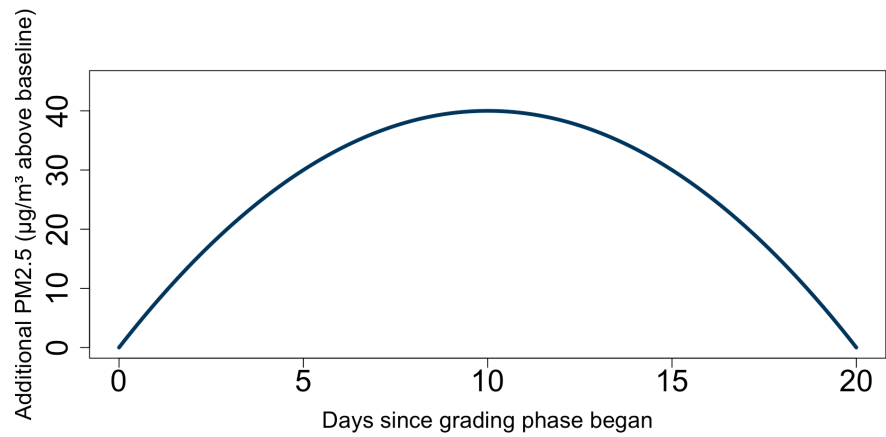
differs from

“Between 1972 and 2020 the price of hobbit homes increased by \$2,450 per year.”

Slope practice

 A city has begun a 20-day, round-the-clock grading and paving phase for a highway-widening project that cuts directly through a neighborhood. In response, a community air-quality monitoring group tracked the additional fine particulate matter concentration, $P(t)$ ($\mu\text{g}/\text{m}^3$ of PM2.5 above baseline), in the neighborhood's air over the 20 days of that construction phase.

t (days)	$P(t)$ ($\mu\text{g}/\text{m}^3$)
2	14.4
6	33.6
10	40.0
14	33.6
18	14.4

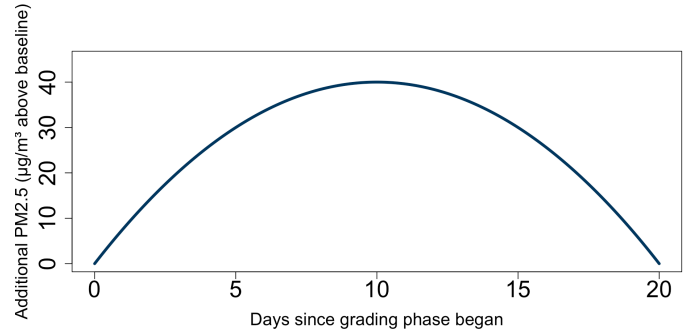


- a. Calculate the average rate of change in additional PM2.5 concentration between day 2 and day 6. Represent this graphically on the plot.
- b. Write your answer as a full sentence, with units, without overstating certainty.

Slope practice

$P(t)$ measures $\mu\text{g}/\text{m}^3$ of PM2.5 above baseline in the neighborhood's air over the 20 days of that construction phase.

t (days)	$P(t)$ ($\mu\text{g}/\text{m}^3$)
2	14.4
6	33.6
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18	14.4



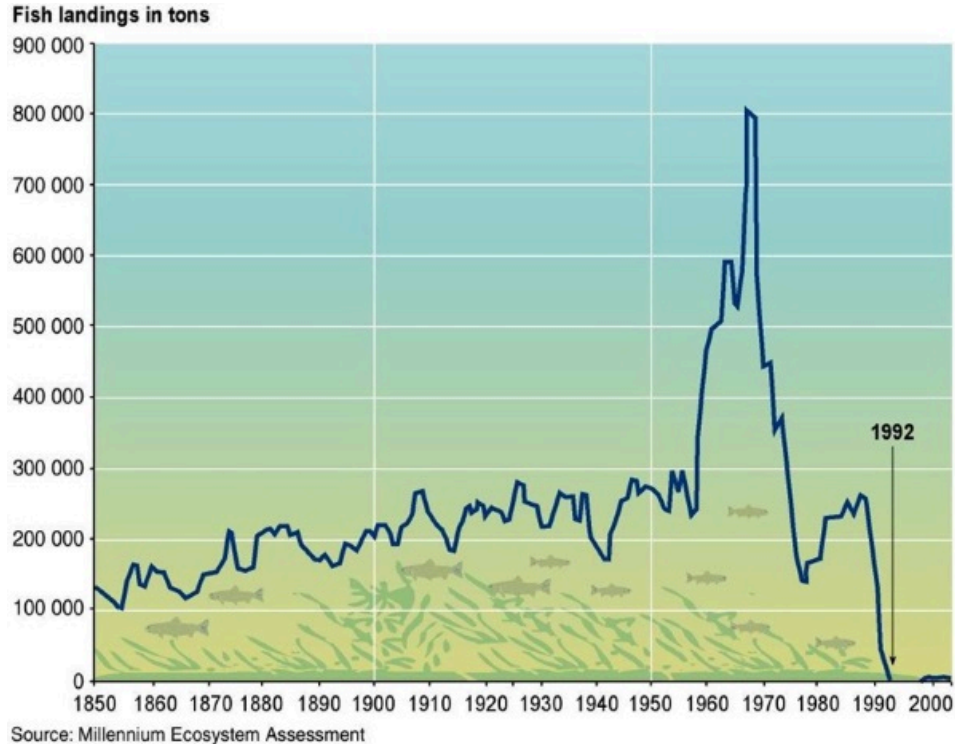
- Calculate the average rate of change in additional PM2.5 concentration between day 2 and day 6. Represent this graphically on the plot.
- Write your answer as a full sentence, with units, without overstating certainty.

Slope practice

a. average rate of change = $\frac{P(6) - P(2)}{6 - 2} = \frac{33.6 - 14.4}{4} = \frac{19.2}{4} = 4.8 \frac{\mu\text{g}/\text{m}^3}{\text{day}}$

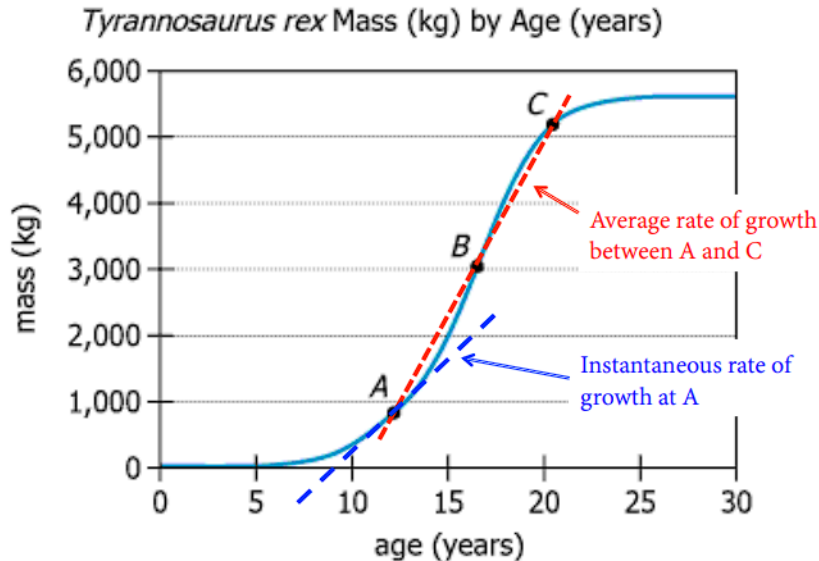
b. Between day 2 and day 6 of the grading phase, additional PM2.5 concentration in the neighborhood increased by an average of 4.8 $\mu\text{g}/\text{m}^3$ per day.

The *average* slope of a continuous function rarely tells the whole story



Slope can be *average* or *instantaneous*

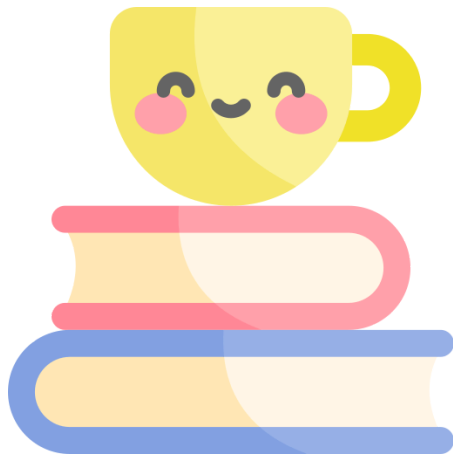
- Rise over run can always be used to find average rate of change between two parts of a graph
- Instantaneous rate of change leads us to ✨calculus✨.



Wrapping up...

Get some extra practice!

Problem set



What we covered today

-  **Graphing**
-  **x and y intercepts**
-  **The number e and $\exp()$**
-  **Exponential growth**
-  **Logistic growth**
-  **Natural logarithms**
-  **Linear functions**
-  **Slope as a rate of change**
-  **Applications from MESM courses**

Logistic growth (extra)

Logistic growth

$$N_t = \frac{K}{1 + \left[\frac{K - N_0}{N_0} \right] e^{-rt}}$$

Where N_t is the population size at time t , K is the carrying capacity, N_0 is the initial population size, and r is a growth rate.



What should be the value of the formula when $t = 0$? Confirm your answer by substituting.

We should always *understand the components of formulas and equations* - both conceptually and mathematically.

Logistic growth: Example

What is

$$N_t = \frac{K}{1 + \left[\frac{K - N_0}{N_0} \right] e^{-rt}}$$

when $t = 0$?

Solution

$$N_0 = \frac{K}{1 + \left[\frac{K-N_0}{N_0} \right] e^{-r(0)}}$$

Substitute $t = 0$

$$= \frac{K}{1 + \left[\frac{K-N_0}{N_0} \right] (1)}$$

$$e^0 = 1$$

$$= \frac{K}{\frac{N_0 + (K - N_0)}{N_0}}$$

Combine terms in the denominator

$$= \frac{K}{\frac{K}{N_0}}$$

$$N_0 + (K - N_0) = K$$

$$= N_0$$

Logistic growth

$$N_t = \frac{K}{1 + \left[\frac{K - N_0}{N_0} \right] e^{-rt}}$$

1. Why might we expect logistic growth for many populations?
2. What variables *besides* time would influence the actual population?